Waseem Raza, PhD (Electrical and Computer Engineering)

ECE Educator | Computer Networking, Data Science & Wireless AI

Undergraduate Engineering Instruction | Computing & Networking | Data & Information Science

✉ waseem.placements@gmail.com | waseemraza844@gmail.com

New Jersey, United States ☎ +1 732-803-9703

LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Academic Profile

PhD in Electrical and Computer Engineering (University of Oklahoma, 2025) with undergraduate teaching experience in computer programming, communication systems and wireless communications. Additional experience includes engineering laboratories, student-project mentoring, and research supervision. Specialization in AI/ML for wireless networks, data-driven network optimization and applied telecommunications engineering through research at the University of Oklahoma and industry work at AT&T Labs, Futurewei and AT&T. Interested in supporting undergraduate lectures, precepts, laboratories, office hours and assessment in computing, networking and data-intensive ECE.

Research Interests

Computer Networking and Wireless Systems; Data and Information Science; Machine Learning for Networked Systems; Network Optimization, Reliability and Resilience; Simulation, Data Analysis and Reproducible Experiments

Doctoral Research Focus

Doctoral research on AI/ML for wireless networks, including data-driven radio propagation modeling, robust learning, generative augmentation, network optimization and self-healing.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*

University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*

University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*

University of Engineering and Technology, Taxila, Pakistan

Relevant Graduate Coursework

LTE Networks and Telecommunications; Emerging 5G Systems; Advanced ML and Computer Vision; Wireless Testbed Design; Numerical Analysis; Information Theory; Digital Signal Processing

Teaching, Research & Professional Experience

Teaching & Instruction

Lecturer, Department of Technology **12/2017 - 09/2019**

University of Lahore, Pakistan *Lahore, Pakistan*

- Taught programming, communication systems, and wireless communication with practical examples.
- Advised senior-design students on projects, academic plans, and early careers.
- Aligned course materials with learning outcomes and accreditation standards.
- Supported accreditation, exams, technical workshops, and student activities.

Teaching areas: Computer Programming; Computer Networking and Wireless Communications; Communication Systems; Data Science and Applied Machine Learning; Signals, Systems and Data Analysis; Laboratory and Project-Based Engineering Instruction

Student mentoring and research development

- Mentored student communications and wireless research projects through simulations and technical presentations.
- Guided student projects and reports with an emphasis on reproducibility and constructive feedback.
- Developed practical labs linking communication theory with engineering applications.

University Research Appointments

Graduate Research Assistant **09/2019 - 07/2025**

AI4Networks, University of Oklahoma *Tulsa, OK*

- Built 3GPP-compliant Python simulation tools and an AI-enabled 5G research testbed.
- Built GAN, transfer-learning, autoencoder, and reinforcement-learning models for wireless data.
- Led 100+ experiments testing wireless ML under domain shift and limited labeled data.
- Developed self-healing and outage-management methods, improving resilience by 30%.
- Published wireless AI research and mentored junior researchers in experimental design.

Research Associate, ComSens Lab **06/2016 - 12/2017**

COMSATS University Islamabad, Pakistan *Islamabad, Pakistan*

- Modeled wireless communication systems through simulation, analysis, and technical reports.
- Mentored Bachelor's and Master's theses in experiment design and research presentation.
- Reviewed student reports and manuscripts for rigor and publication readiness.
- Supported proposals, literature reviews, lab seminars, and research coordination.

Full-Time Research Scholar, Communication Systems Lab **08/2014 - 02/2016**
University of Engineering and Technology, Taxila, Pakistan *Taxila, Pakistan*

- Researched wireless communication systems while completing an MSc in Telecommunications.
- Supported communications courses, labs, assignments, grading, and examinations.
- Led wireless communication labs and prepared technical manuals and experiments.
- Developed course and student-project materials for practical wireless engineering.

Industry Research & Engineering

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Applied domain-adaptive ML to telecom service assurance and model validation.
- Designed cross-domain evaluation and error-analysis methods for operational ML.
- Worked with engineering and public-safety teams on reliable network assurance.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built LLM-enabled analytics to interpret geochemical well data for decision support.
- Built Python pipelines in Azure Databricks, Functions, and Synapse Analytics.
- Automated exploration-data ETL, cutting preparation time by 40%.
- Used data-validation and quality controls to improve output reliability by 20%.
- Turned geologists' technical needs into structured datasets and useful analytics.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR logs with ML, reducing diagnostic time by 20%.
- Improved 5G NR-NTN cell selection by 15% through data-driven validation.
- Presented operational data insights to next-generation wireless engineering teams.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved wireless CSI compression by 25% with transfer learning and autoencoders.
- Ran 50+ experiments on wireless ML robustness under changing data conditions.
- Translated research results into recommendations for future wireless networks.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Improved 5G standalone user-timing precision by 20% with ML and statistics.
- Analyzed user traces with SQL and presented results to engineering leadership.
- Turned operational traces into indicators for network diagnosis and optimization.

Research Contributions

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *HealthINF | Accepted* 2026

AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks *Computer Communications* 2025 | DOI

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *Sensors 2025* 2025 | DOI

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *IEEE PIMRC* 2024 | DOI

Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity *IEEE VTC-Fall 2024* 2024 | Link

An AI-Enabled Framework to Defend Ingenious MDT-Based Attacks on the Emerging Zero Touch Cellular Networks *IEEE Network, 38(1), 228-237* 2024 | DOI

Positioning Error Impact Compensation through Data-Driven Optimization in User-Centric Networks *IEEE GLOBECOM 2023* 2023 | Link

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *IEEE Network* 2022 | Link

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *IEEE WCNC* 2022

Research Projects & Platforms

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

Research Methods & Technical Expertise

Undergraduate Teaching & Course Support: Computer programming, communication systems and wireless communications instruction; communications laboratories, student mentoring, course materials, technical discussions, assessment and feedback

Computer Networking & Wireless Systems: LTE/4G, 5G-NR, radio access networks, wireless communications, network measurements, propagation, network reliability, service assurance, self-healing and optimization

Data & Information Science: Python, SQL, data preprocessing, statistical analysis, machine learning, classification, regression, anomaly detection, time-series analysis, evaluation and visualization

AI/ML & Network Modeling: Deep learning, generative modeling, transfer learning, reinforcement learning, distribution-shift robustness, data-driven simulation and predictive optimization

Programming & Research Tools: Python, C/C++, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Git, Linux and reproducible computational workflows

Academic Service & Recognition

Academic Service

- Developed course materials and supported curriculum assessment and accreditation activities.
- Supported research seminars, engineering laboratory instruction and student mentoring.

Honors & Awards

William H. Barkow Scholarship *2024*
University of Oklahoma, Electrical and Computer Engineering

Clyde L. Farrar Endowment Scholarship *2024*
University of Oklahoma

P.H. Robinson Fellowship *2023*
University of Oklahoma

FAS-Charles Forgan Scholarship *2023*
University of Oklahoma

Self-Learning

1. Machine Learning Stanford/Coursera | Completed | Mar 2020
Supervised and unsupervised learning, regularization, model evaluation, and practical predictive modeling.

2. Python for Data Science, AI & Development IBM/Coursera | Completed | Jul 2026
Python programming, data analysis, APIs, and reproducible data-science workflows.

3. AI Foundations & Industry Overview for Telecommunication Coursera | Completed | Jun 2026
AI applications in telecom networks, operations analytics, service assurance, and automation.

4. Network and Security Optimization in Telecommunication Coursera | Completed | Jun 2026
Data-driven network optimization, telecom security, reliability analysis, and operational efficiency.

5. 4G Network Fundamentals Mines-Télécom/Coursera | Completed | Aug 2026
LTE radio and core networks, cellular architecture, mobility management, and 4G fundamentals.